🩺 VisitPrep

DSH Hacks V2 · AI × Healthcare · Pierre Arthur Tchio Seme (SAIT, Calgary)

Problem

Primary-care visits are often 10–15 minutes. Patients struggle to describe symptoms clearly and forget key details such as when symptoms began, how severe they are, and what they have already tried. Incomplete histories slow care and frustrate both sides.

Solution

VisitPrep is a web app where patients describe how they feel in their own words. An on-device natural-language-processing engine converts that story into a structured, doctor-ready summary and a list of tailored questions to ask.

Key features

- **Free-text symptom extraction** across 16 symptom groups with lay synonyms ("head hurts" → headache)
- **Negation handling**: "no fever" is not recorded as fever
- **Duration, severity, pattern and medication detection** ("for 3 days", "really bad", "worse in the morning", "Tylenol")
- **Emergency red-flag detection** (chest pain, trouble breathing, stroke signs, self-harm) with a clear call-911 prompt
- **Doctor-ready summary + suggested questions**, one tap to copy
- **Privacy by design**: runs entirely in the browser; nothing is uploaded

How it works

Static HTML/CSS/JavaScript. `nlp.js` implements a transparent rule-based pipeline: lexicon matching → negation window check → regex extraction of duration, severity, modifiers and medications → red-flag screening. `app.js` renders the summary and question list.

Impact

Helps patients, especially seniors, newcomers and anxious patients, communicate better, and gives clinicians a consistent snapshot. It costs nothing to run and works on any phone.

Next steps

On-device ML model for richer extraction, multilingual input, clinic intake-form export, and evaluation with clinicians.

VisitPrep is a prototype and does not diagnose or give medical advice.